

① $\exists f \in C^0[A, B]$, $x = \varphi(t): [\alpha, \beta] \rightarrow [A, B]$,

$\varphi \in \Phi[\alpha, \beta]$, $\varphi' \in R_{loc}[\alpha, \beta]$,

$\exists \lim_{t \rightarrow \beta-0} \varphi(t) \in \overline{\mathbb{R}} \Rightarrow$

$$\int_{\varphi(\alpha)}^{\varphi(\beta-0)} f(x) dx = \int_{\alpha}^{\beta} f(\varphi(t)) \varphi'(t) dt,$$

к тому же, если $\exists$ open neighborhood of $\overline{\mathbb{R}}$, то

$\exists$ unique

$$\Phi(x) = \int_{\alpha}^x f(\varphi) \varphi' dt$$

$$F(C) = \int_{\varphi(\alpha)}^{\varphi(\beta-0)} f dx$$

т.к. $\Phi(x) = F(\varphi(x))$, $x \in [\alpha, \beta]$

$$\textcircled{1} \exists \int_{\varphi(\alpha)}^{\varphi(\beta-0)} f dx = I \in \overline{\mathbb{R}}.$$

$\exists \gamma_n \in [\alpha, \beta]: \{\gamma_n\} \xrightarrow{n \rightarrow \infty} \beta \Rightarrow \begin{cases} \varphi(\gamma_n) \in [A, B] \\ \{\varphi(\gamma_n)\} \xrightarrow{n \rightarrow \infty} \varphi(\beta-0) \end{cases}$

$$\Rightarrow \lim_{n \rightarrow \infty} \Phi(\gamma_n) = \lim_{n \rightarrow \infty} F(\varphi(\gamma_n)) = I$$

$$\Rightarrow \text{т.к. } \{\gamma_n\} - \forall \Rightarrow \int_{\alpha}^{\beta} f(\varphi) \varphi' dt = I \in \overline{\mathbb{R}}$$

$$\textcircled{2} \exists \int_{\alpha}^{\beta} f(\varphi) \varphi' dt = I \in \overline{\mathbb{R}}$$

$$a) \exists \varphi(\beta-0) \in [A, B] \Rightarrow \int_{\varphi(\alpha)}^{\varphi(\beta-0)} f dx = \int_{\varphi(\alpha)}^{\varphi(\beta-0)} f dx - \int_{\varphi(\alpha)}^{\varphi(\beta-0)} f dx$$

$$\Rightarrow \text{н.у. } \int_{\varphi(\alpha)}^{\varphi(\beta-0)} f dx = I \in \mathbb{R}.$$

$$b) \exists \varphi(\beta-0) = B \quad \exists \{C_n\} \in [A, B]$$

$$\{C_n\} \xrightarrow{n \rightarrow \infty} B$$

$$\exists C_n \in [\varphi(\alpha), B] \Rightarrow$$

$$\exists \gamma_n \in [\alpha, \beta) : \varphi(\gamma_n) = C_n \Rightarrow \{\gamma_n\} \xrightarrow{n \rightarrow \infty} \beta ?$$

$$\exists \{\gamma_{n_k}\} \xrightarrow{k \rightarrow \infty} \tau, \quad \tau \in [\alpha, \beta) \Rightarrow$$

$$\varphi(\gamma_{n_k}) \xrightarrow{k \rightarrow \infty} \varphi(\tau) < \beta \quad (\text{т.к. } \varphi \in C^0)$$

а это неверно! (т.к. $\varphi(\beta-0) = \beta$ по условию)

$$\Rightarrow \{\gamma_n\} \xrightarrow{n \rightarrow \infty} \beta \Rightarrow \lim_{n \rightarrow \infty} F(C_n) = \lim_{n \rightarrow \infty} \Phi(\gamma_n) = \bar{I} \quad \Delta$$

Реш. $\int_0^{\pi} \frac{dx}{2 + \cos x} = \left\{ \begin{array}{l} t = \operatorname{tg} \frac{x}{2} \\ x = 2 \operatorname{arctg} t \\ x=0, t=0 \\ x=\pi, t=+\infty \end{array} \right. \quad \left. \begin{array}{l} dx = \frac{2 dt}{1+t^2} \\ \cos x = \frac{1-t^2}{1+t^2} \end{array} \right\}$

$$= 2 \int_0^{+\infty} \frac{dt}{(1+t^2)(3+t^2)} = 2 \int_0^{+\infty} \frac{dt}{3+t^2} = \frac{2}{\sqrt{3}} \operatorname{arctg} t \Big|_0^{+\infty} =$$

$$= \frac{2}{\sqrt{3}} \frac{\pi}{2} = \frac{\pi}{\sqrt{3}}.$$

Зам. $x = b - \frac{1}{t} : \left[\frac{1}{b-a}, +\infty \right) \rightarrow [a, b)$

$$\Rightarrow \int_a^b f(x) dx = \int_{\frac{1}{b-a}}^{+\infty} f\left(b - \frac{1}{t}\right) \frac{dt}{t^2},$$

т.е. $\int_{\frac{1}{b-a}}^{+\infty} f\left(b - \frac{1}{t}\right) \frac{dt}{t^2}$

① Критерий сходимости интеграла от знакопеременной функции.

$$\exists f \in R_{loc} [a, b), \quad \underline{f \geq 0} \Rightarrow$$

$$F(\omega) = \int_a^{\omega} f dx, \quad \omega \in [a, b) - \text{базис.}$$

$$\int_a^b f dx \text{ — сс — сл} \Leftrightarrow F(\omega) - \underline{\text{о.ч.}}$$

$$\Delta \quad \exists w_1, w_2 : a \leq w_1 < w_2 < b, \quad f > 0$$

$$\Rightarrow \int_{w_1}^{w_2} f dx > 0 \quad \Rightarrow$$

$$\circ \quad F(w_2) = \int_a^{w_2} f dx = \int_a^{w_1} f dx + \int_{w_1}^{w_2} f dx > 0$$

$$\Rightarrow \int_a^{w_1} f dx = F(w_1) \Rightarrow F(w_2) > F(w_1).$$

$$\textcircled{H} \quad \exists \int_a^b f dx - cx \Rightarrow \exists \lim_{w \rightarrow b-0} F(w) \in \mathbb{R}$$

$$\Rightarrow F(w) - \text{огр} \quad \text{на } [a, b].$$

$$\textcircled{I} \quad \exists F(w) - \text{огр} \quad \text{на } [a, b]$$

$$\Rightarrow \text{T. k. } \uparrow F(w) - \text{огр} \xrightarrow{T \beta - \text{огр}} \Rightarrow \exists \lim_{w \rightarrow b-0} F(w) \in \mathbb{R}$$

$$\Rightarrow \int_a^b f dx - cx. \quad \Delta$$

Т Признаки сравнения.

$$\exists f, g \in R_{\text{огр}} [a, b]$$

$$\underline{0 \leq f \leq g, \quad x \in [a, b]} \quad (\Rightarrow)$$

$$\textcircled{1} \quad \text{если } \int_a^b g dx - cx, \text{ то } \int_a^b f dx - cx$$

$$\textcircled{2} \quad \text{если } \int_a^b f dx - p/cx, \text{ то } \int_a^b g dx - p/cx$$

$$\textcircled{3} \quad \text{если } f \sim g \text{ при } \underline{x \rightarrow b-0} \Rightarrow \text{оба интеграла сходятся или расходятся}$$

$$\textcircled{4} \quad \text{если } \exists \lim_{x \rightarrow b-0} \frac{f}{g} = l \quad \left(\begin{array}{l} l \neq 0 \\ l \neq \infty \end{array} \right) \Rightarrow \text{выполнено - то.}$$

$$\Delta \quad 1) \quad F(w) = \int_a^w f dx - \text{не убывает с ростом } w$$

$$\Rightarrow F(w) = \int_a^w f dx \leq \int_a^w g dx \leq \sup_{w \in [a, b]} \int_a^w g dx = \int_a^b g dx \in \mathbb{R}$$

$$\Rightarrow \int_a^b f dx - cx - \text{огр} \quad (\text{но } T \uparrow \text{ и } T \beta - \text{огр})$$

2) ст. н. прот-но: $\int_a^b g dx = cn \xrightarrow{(1)} \int_a^b f dx = cn.$
 а ст. н. прот-но с условием. •

3) $\int f \sim g, x \rightarrow b-0 \Rightarrow \exists \alpha(x):$

$$\begin{cases} f(x) = \alpha(x) \cdot g(x) & x \in U(b) \cap [a, b) \\ \lim_{x \rightarrow b-0} \alpha(x) = 1 \end{cases}$$

$$\Rightarrow \forall \varepsilon > 0 \exists \delta = \delta(\varepsilon) > 0: \forall x \in U_\delta(b) \Rightarrow$$

$$\Rightarrow |\alpha(x) - 1| < \varepsilon \Rightarrow$$

$$1 - \varepsilon < \alpha(x) < 1 + \varepsilon$$

$$(1 - \varepsilon) g(x) < \alpha(x) g(x) < (1 + \varepsilon) g(x)$$

$$(1 - \varepsilon) g(x) < f(x) < (1 + \varepsilon) g(x) \quad (*)$$

$$\int_a^b f dx = cn \xrightarrow{(1)} \int_a^b (1 - \varepsilon) g dx = cn$$

$$\Rightarrow \int_a^b g dx = cn.$$

$$\int_a^b f dx = p/cn \Rightarrow \int_a^b (1 + \varepsilon) g dx = p/cn$$

$$\Rightarrow \int_a^b g dx = p/cn. \text{ н.р.г. } \triangle$$

(4) γ_{np}

Ип $\int_1^{+\infty} \frac{x dx}{\sqrt[3]{1 + 2x^2 + x^5}}$

$$f(x) > 0$$

$$\begin{aligned} \frac{x}{\sqrt[3]{1 + 2x^2 + x^5}} &= \frac{x}{x^{5/3} \left(\frac{1}{x^5} + 2 \frac{x^2}{x^5} + 1 \right)^{1/3}} \sim \frac{x}{x^{5/3}} = \\ &= \frac{1}{x^{2/3}} \end{aligned}$$

т.к. $\int_1^{+\infty} \frac{dx}{x^{2/3}} = p/cx \rightarrow 0 \quad \left[\frac{2}{3} < 1 \right]$

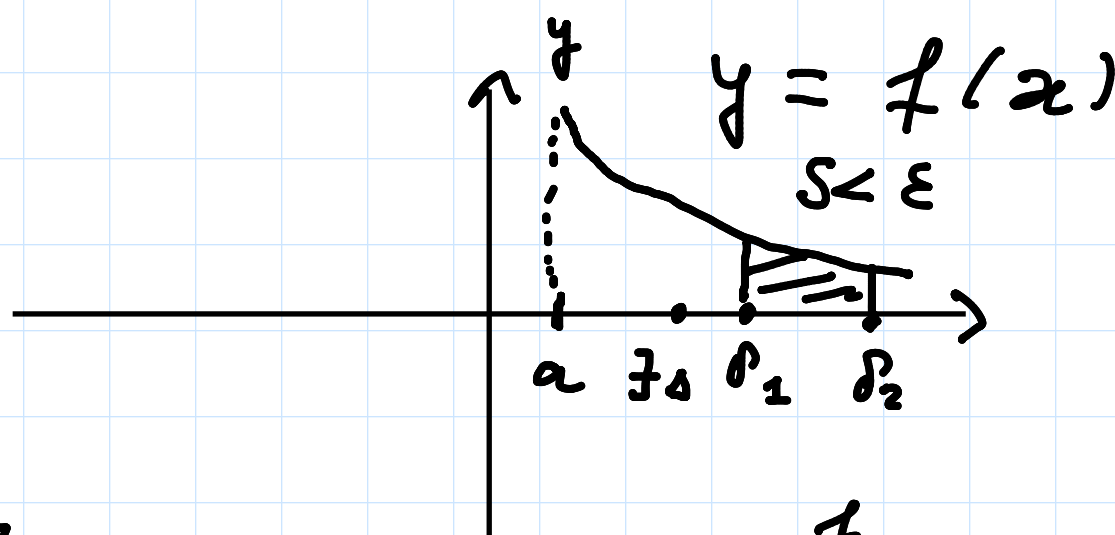
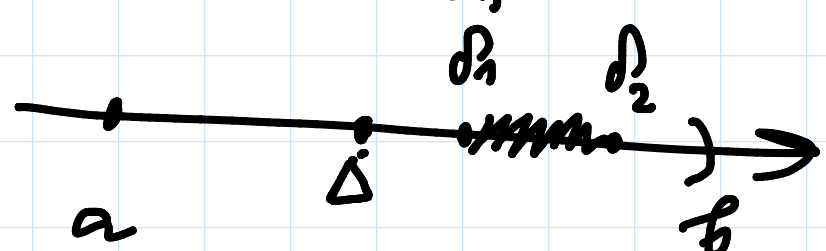
$\Rightarrow$ по пр. ср. п/сх и мсх. интервал.

⑦ Критерий Коши сх-ти и.и.

$\exists f \in R_{loc} (a, b)$, $\int_a^b f dx$ - сс. $(\Leftrightarrow)$

$\forall \varepsilon > 0 \exists \Delta \in (a, b) : \forall \rho_1, \rho_2 \in (\Delta, b)$

$$\Rightarrow \left| \int_{\rho_1}^{\rho_2} f dx \right| < \varepsilon$$



$\Delta \exists F(\omega) = \int_a^\omega f dx$, $\exists \int_a^b f dx$ - сс.

$\Rightarrow \exists \lim_{\omega \rightarrow b-0} F(\omega) \in \mathbb{R}$ $(\Leftrightarrow)$ крк где Φ -члн $F(\omega)$

$\forall \varepsilon > 0 \exists \Delta \in (a, b) :$

$\forall \rho_1, \rho_2 \in (\Delta, b) \Rightarrow |F(\rho_2) - F(\rho_1)| < \varepsilon$

$$\Rightarrow \left| \int_{\rho_1}^{\rho_2} f dx \right| < \varepsilon \quad \Delta$$

Отрицание критерия Коши.

$\exists f \in R_{loc} (a, b)$, $\int_a^b f dx$ - п/сх $(\Leftrightarrow)$

$\exists \varepsilon > 0 : \forall \Delta \in (a, b) \exists \rho_1, \rho_2 \in (\Delta, b) :$

$$\left| \int_{\rho_1}^{\rho_2} f dx \right| > \varepsilon$$

$\int_{\mathbb{R}} \frac{\sin x}{x} dx$

$\rho_1 = \pi n$
 $\rho_2 = 2\pi n$!

$\Rightarrow \left| \int_{\pi n}^{2\pi n} \frac{\sin x}{x} dx \right| > \frac{1}{2\pi n} \int_{\pi n}^{2\pi n} |\sin x| dx =$

$$= \frac{1 \cdot n}{2\pi \cdot n} \int_0^{2\pi} |\sin x| dx = \frac{1}{2\pi} \cdot 2 = \frac{1}{\pi}$$

$$= -\cos x \Big|_0^{2\pi}$$

$$\exists \varepsilon = \frac{1}{\pi} : \forall \Delta \in [1, +\infty) \exists \delta_1 = \pi n, \delta_2 = 2\pi n :$$

$$\left| \int_{\delta_1}^{\delta_2} f dx \right| > \frac{1}{\pi} \xRightarrow{\text{кр.к.}} \text{неч. и. н. } f|_{\cos}$$

Опр.

$$\exists f \in R_{\text{loc}} [a, b)$$

и. н. от. f на $[a, b)$ нег. абсолютно сж-е,
если сж. $\int_a^b |f| dx$

$$\textcircled{T} \exists f \in R_{\text{loc}} [a, b), \exists \int_a^b f dx - \text{сж. сжс.} \Rightarrow$$

$$\Rightarrow \int_a^b f dx - \text{сж.}$$

$$\Delta \exists \int_a^b f dx - \text{сж. сжс.} \Rightarrow \int_a^b |f| dx - \text{сж}$$

кр.к.

$$\Rightarrow \forall \varepsilon > 0 \exists \Delta : \forall \delta_1, \delta_2 \in (\Delta, b) \Rightarrow$$

$$\left| \int_{\delta_1}^{\delta_2} |f| dx \right| < \varepsilon \Rightarrow$$

$$\left| \int_{\delta_1}^{\delta_2} f dx \right| \leq \left| \int_{\delta_1}^{\delta_2} |f| dx \right| < \varepsilon$$

$$\xRightarrow{\text{кр.к.}} \int_a^b f dx - \text{сж.} \quad \Delta$$

$$\text{Пр.} \int_1^{+\infty} \frac{\sin x}{x^2} dx, \quad \int_1^{+\infty} \left| \frac{\sin x}{x^2} \right| dx$$

$$\left| \frac{\sin x}{x^2} \right| \leq \frac{1}{x^2} \quad \int_1^{+\infty} \frac{dx}{x^2} - \text{сж}$$

$$\Rightarrow \int_1^{+\infty} \left| \frac{\sin x}{x^2} \right| dx - \text{сж.} \Rightarrow \text{сж. и. н. сж. сжс.}$$

Опр. $f \in \mathcal{R}_{loc} [a, b)$

Есть

$$\left\{ \begin{array}{l} \int_a^b f \, dx = c \\ \int_a^b |f| \, dx = p < \infty \end{array} \right. \Rightarrow \int_a^b f \, dx = \underline{c.} \\ \text{условно}$$

Пр.

$\int_1^{+\infty}$

$$\frac{\sin x}{x} \, dx$$

$= c. \text{ условно?}$